

Activity

1) Medical Image Enhancement.

1) Which gray-level transformation would you recommend?

A: Log Transformation.

2) Why is it suitable?

A: Log transformation enhances the darkest regions of the X-ray image without greatly affecting the brightened bone regions.

3) What effect will it have on the image?

A: The dark tissues become brighter and cleared improving visibility, while the bones remain visible with better overall contrast.

2) Night - Time Surveillance.

1. Which transformation would improve the visibility?

Log Transformation.

2) Why is it more suitable than negative transformation?

Log transformation increases the brightness of dark pixels, making objects easier to see.

⇒ Negative transformation only inverts the image colors and does not improve visibility effectively.

3) The image will vehicles, people, and other processing important data.

3) Fingerprint Recognition

1) Which gray-level

A: Contrast stretching.

2) Explain why.

A: Contrast stretch the ridges and sharpen and edge.

3) What would applied instead

Log transformation but would as effective

3) The image will appear brighter, with roads, vehicles, people, and others becoming more visible while preserving important details.

3) Fingerprint Recognition.

1) Which gray-level transformation would you apply?

A. Contrast stretching.

2) Explain why.

A. Contrast stretching increases the difference b/w the ridges and valleys, making fingerprint patterns sharper and easier to recognize.

3) What would happen if a log transformation were applied instead?

Log transformation would brighten the darker regions but would not improve the ridge-to-valley contrast as effectively as contrast stretching.